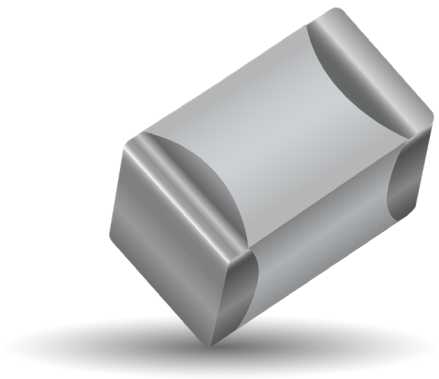
RF/Microwave Capacitors

RF/Microwave Multilayer Capacitors (MLC)

800B Series NP0 Porcelain, High RF Power Ultra-Low ESR



GENERAL DESCRIPTION

KYOCERA AVX's 800B Series offers superb performance in demanding high RF power applications requiring consistent and reliable operation. The combination of highly conductive metal electrode systems, optimized case geometries, and proprietary dielectrics, yields the lowest ESR. KYOCERA AVX's new NPO low loss rugged dielectrics are designed to provide superior heat transfer in high RF power applications. Ultra-low ESR and superior thermal performance insure that the 800B Series products are your best choice for high RF power applications from VHF through microwave frequencies.

TYPICAL APPLICATIONS

- Avionics
- Public Safety Radio
- Wireless Communications
- VHF / UHF / HDTV Broadcast Transmitters
- Telecom
- WiMAX
- Microwave Communication Systems and Satellite Systems

TYPICAL CIRCUIT APPLICATIONS

- High RF Power Filter Networks
- Matching Networks
- Couplers
- Combiners
- Output Coupling
- Antenna Coupling
- DC Blocking
- Bypassing

ENVIRONMENTAL TEST

Thermal Shock	MIL-STD-202, Method 107, Condition A
Moisture Resistance	MIL-STD-202, Method 106
Low Voltage Humidity	MIL-STD-202, Method 103, Condition A, with 1.5 Volts DC applied while subjected to an environment of 85°C with 85% relative humidity for 240 hours min.
Life Test	MIL-STD-202, Method 108, for 2000 hours, at 125°C 200% WVDC applied

FEATURES

- Case B Size (.110" x .110")
- Rugged, reliable NPO dielectric
- Case optimized for highest self resonant frequency
- Capacitance Range 0.1 pF to 1000 pF
- Lowest ESR
- Capable of highest RF Power
- RoHS Compliant/Lead-Free

PACKAGING OPTIONS

Tape & Reel

Vertical Orientation Tape & Reel

RoHS COMPLIANT

ENVIRONMENTAL CHARACTERISTICS

Quality Factor	Greater than 2,000 at 1 MHz (Cap Values <=10pf .0010 Max. @ 1MHz Cap Values >10pf .0005 Max. @ 1MHz)
Temperature Coefficient of Capacitance (TCC)	0 ±30 PPM/°C (-55°C to +125°C)
Insulation Resistance (IR)	0.1 pF to 1000 pF: 10^5 Megohms min. @ +25°C at rated WVDC 10^4 Megohms min. @ +125°C at rated WVDC
Working Voltage (WVDC)	See Capacitance Values Table
Dielectric Withstanding Voltage (DWV)	Case B: 250% of rated WVDC for 5 secs
Retrace	Less than ±(0.02% or 0.02 pF), whichever is greater
Aging Effects	None
Piezoelectric Effects	None
Capacitance Drift	±(0.02% or 0.02 pF), whichever is greater
Operating Temperature Range	From -55°C to +125°C (No derating of working voltage)
Termination Styles	See Mechanical Configurations
Terminal Strength	Terminations for chips withstand a pull of 5 lbs. min., 15 lbs. typical, for 5 seconds in direction perpendicular to the termination surface of the capacitor. Test per MIL-STD-202, method 211.

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CAPACITANCE VALUES

Cap. Code	Cap. (pF)	Tol.	Rated WVDC	Cap. Code	Cap. (pF)	Tol.	Rated WVDC	Cap. Code	Cap. (pF)	Tol.	Rated WVDC	Cap. Code	Cap. (pF)	Tol.	Rated WVDC
0R1	0.1	B	500	2R4	2.4	B, C, D	500	200	20	F, G, J, K, M	500	151	150	F, G, J, K, M	300
0R2	0.2			2R7	2.7			220	22			161	160		
0R3	0.3	B, C		3R0	3.0			240	24			181	180		
0R4	0.4			3R3	3.3			270	27			201	200		
0R5	0.5	B, C, D		3R6	3.6			300	30			221	220		200
0R6	0.6			3R9	3.9			330	33			241	240		
0R7	0.7			4R3	4.3			360	36			271	270		
0R8	0.8			4R7	4.7			390	39			301	300		
0R9	0.9			5R1	5.1			430	43			331	330		
1R0	1.0			5R6	5.6			470	47			361	360		
1R1	1.1			6R2	6.2			510	51			391	390		
1R2	1.2			6R8	6.8	B, C, J, K, M		560	56			431	430		
1R3	1.3			7R5	7.5			620	62			471	470		
1R4	1.4			8R2	8.2			680	68			511	510		100
1R5	1.5			9R1	9.1			750	75			561	560		
1R6	1.6			100	10	F, G, J, K, M		820	82			621	620		
1R7	1.7			110	11			910	91			681	680		50
1R8	1.8			120	12			101	100			751	750		
1R9	1.9			130	13			111	110		300	821	820		
2R0	2.0			150	15			121	120			911	910		
2R1	2.1			160	16			131	130			102	1000		
2R2	2.2			180	18										

VRMS = 0.707 X WVDC
• SPECIAL VALUES, TOLERANCES AND MATCHING AVAILABLE. PLEASE CONSULT FACTORY.

HOW TO ORDER

800 B 910 J T 500 X T

Series

Case Size
See mechanical dimensions below

Capacitance
EIA Capacitance Code in pF.
First two digits = significant figures or "R" for decimal place.
Third digit = number of zeros or after "R" significant figures

Capacitance Tolerance Code

Code	B	C	D	F	G	J	K	M
Tol.	±0.1 pF	±0.25 pF	±0.5 pF	±1%	±2%	±5%	±10%	±20%

Packaging
T = Tape and Reel, 500 pc qty
TV = Vertical Tape and Reel, 500 pc qty
Please see last Column Mechanical Configuration Table for other options

Laser Marking (Optional)

WVDC

Termination Code
Please see 2nd Column Mechanical Configuration Table

The above part number refers to a 800 B Series (case size B) 91 pF capacitor, J tolerance (±5%), 500 WVDC, with T termination (Tin Plated over Nickel Barrier Termination, RoHS Compliant), laser marking and tape and reel packaging.

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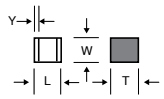
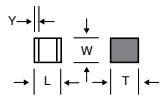
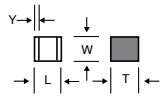
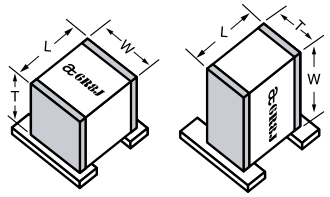
MECHANICAL CONFIGURATION

Series & Case Size	Term. Code	Case Size & Type	Outline ES W/T is a Termination Surface	Body Dimensions inches (mm)			Lead and Termination Dimensions and Material		Pkg Type & Qty	Pkg Code
				Length (L)	Width (W)	Thickness (T)	Overlap (Y)	Materials		
800B	T	B Solderable Nickel Barrier		.110+.020 -.010 (2.79+0.51-0.25)	.110 ±.015 (2.79 ±0.38)	.070 (1.78) max.	.015 (0.38) ±.010 (0.25)	***RoHS Compliant*** Tin Plated over Nickel Barrier Termination	T&R, 1000 or 500 pcs Vertical T&R, 1000 or 500 pcs	T1K or T TV1K or TV
800B	W	B Solder Plate		.110+.020 -.010 (2.79+0.51-0.25)	.110 ±.015 (2.79 ±0.38)			Tin/ Lead, Solder Plated over Nickel Barrier Termination	T&R, 1000 or 500 pcs Vertical T&R, 1000 or 500 pcs	T1K or T TV1K or TV

NON-MAGNETIC CONFIGURATION

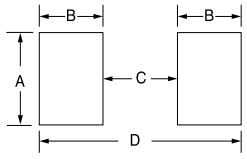
Series & Case Size	Term. Code	Case Size & Type	Non-Magnetic Configuration	Body Dimensions inches (mm)			Lead and Termination Dimensions and Material		Pkg Type & Qty	Pkg Code
				Length (L)	Width (W)	Thickness (T)	Overlap (Y)	Materials		
800B	TN	B Non-Mag Solderable Barrier		.110+.020 -.010 (2.79+0.51-0.25)	.110 ±.015 (2.79 ±0.38)	.070 (1.78) max.	.015 (0.38) ±.010 (0.25)	***RoHS Compliant*** Tin Plated over Non-Magnetic Barrier Termination	T&R, 1000 or 500 pcs Vertical T&R, 1000 or 500 pcs	T1K or T TV1K or TV

SUGGESTED MOUNTING PAD DIMENSIONS



Horizontal Electrode Orientation

Vertical Electrode Orientation



Case B Vertical Mount					
Cap Value .43	Pad Size	A Min.	B Min.	C Min.	D Min.
All Values	Normal	.090 (2.29)	.050 (1.27)	.075 (1.91)	.175 (4.45)
	High Density	.070 (1.78)	.030 (.762)	.075 (1.91)	.135 (3.43)

inches (mm)

Case B Horizontal Mount					
Cap Value .43	Pad Size	A Min.	B Min.	C Min.	D Min.
All Values	Normal	.130 (3.30)	.050 (1.27)	.075 (1.91)	.175 (4.45)
	High Density	.110 (2.79)	.030 (.762)	.075 (1.91)	.135 (3.43)

inches (mm)

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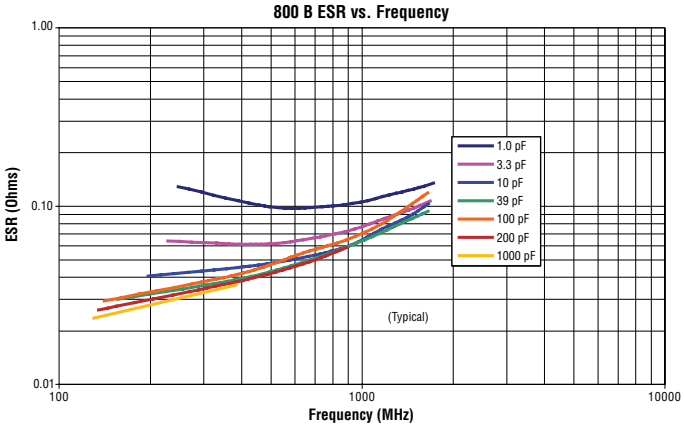
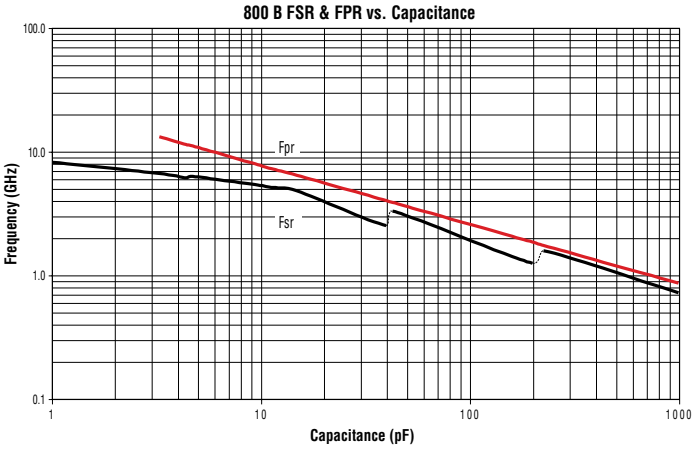
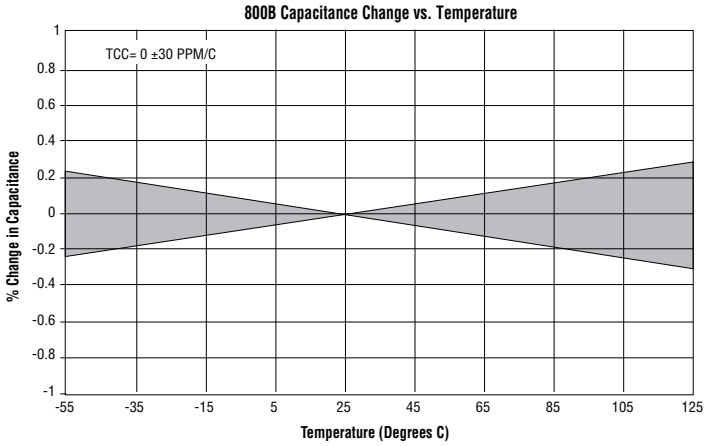
PERFORMANCE DATA







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SAMPLE KITS

Kit #	RoHS Compliant	Item Number	Description	Cap. Value Range (pF)	Cap Value (pF)	Tol.	Price
Kit 83T	RoHS COMPLIANT	DK0083T	**800B NPO Ceramic High RF Power MLCs** 16 different values, 15 pcs. min. per value	1.0 to 10	1.0, 1.2, 1.5, 1.8, 2.0, 2.2, 2.4, 2.7, 3.0, 3.3 3.9, 4.7, 5.6, 6.8, 8.2 10	±0.1 ±0.25 ±5%	$165.00
Kit 84T	RoHS COMPLIANT	DK0084T	**800B NPO Ceramic High RF Power MLCs** 16 different values, 15 pcs. min. per value	10 to 100	10, 12, 15, 18, 20, 22, 24, 27, 30, 33, 39, 47, 56, 68, 82, 100	±0.1	$165.00
Kit 85T	RoHS COMPLIANT	DK0085T	**800B NPO Ceramic High RF Power MLCs** 16 different values, 15 pcs. min. per value	100 to 1000	100, 120, 150, 180, 200, 220, 240, 270, 300, 330, 390, 470 560, 680, 820, 1000	±5% ±10%	$165.00

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